

Suzy Zhou

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 xiaoyun.zhou27@gmail.com, Target: staff/senior research scientist on ML/AI/LLM

WORK & EDUCATION

Senior applied scientist - Amazon USA	2024–now
role: lead the design and development of machine learning systems for non-compliant product detection, collaborating cross-functionally with operations, science, data, seller appeal, and software engineering teams.	
Applied scientist - Amazon USA	2022–2024
projects: multi-modality machine learning for non-compliant product detection	
Senior research scientist - PAII Inc.(PingAn Tech.) USA	2020–2022
projects: deep/machine learning, computer vision for cancer diagnosis	
PhD - Department of computing, Imperial College London, UK	2015–2020
research: deep/machine learning, computer vision, medical robotics and graph learning	
MRes - Medical robotic navigation, Imperial College London, UK	2014–2015
Master - Biomedical engineering, Tsinghua University, China	2011–2014
Bachelor - Comm. eng., Wuhan University of Tech., China, GPA 3.5/4.0	2007–2011

PROJECT EXPERIENCE

Non-compliant product detection

- lead ML system design and technical strategy for detecting non-compliant products across a multi-disciplinary organization, protecting customer experience at scale.
- partner cross-functionally with ops, seller appeal, legal, rule writing, and software engineering teams to build scalable, ML-powered compliance workflows.
- drove model evolution/delivery from traditional CNNs (e.g., ResNet, Swin Transformer) to multi-modal models combining image and text, and further to LLMs capable of understanding full product pages with ambiguous compliance risks (e.g., misleading claims).
- deep familiarity with AWS services and ML infrastructure, including LLM deployment, scalable cloud-based pipelines, and multi-agent system
- acted as technical reviewer, decision-maker, conflict resolution on research direction, model architecture, and system integration for all science efforts in the organization.

Deep/machine learning and computer vision:

- BMD estimation from plain X-rays [1, 4, 8, 24] (delivered), liver tumour identification from CT/MRI images [22, 23], adversarial learning [2, 3, 6], pre-operative image/volume segmentation in both 2D and 3D [11, 12, 15, 18], breast tumour detection from mammograms (delivered), high-value customer mining based on multi-modality data, including video, audio, text and meta data.

Robotic vision:

- intra-operative 3D navigation with real-time 3D shape instantiation from a single intra-operative 2D projection [5, 9, 10, 13], 3D robotic path planning towards surgical autonomy [7, 16, 20, 21].

Graph learning:

- 3D mesh/deformation reconstruction from a single 2D image [14, 17, 24].

SKILLS

<i>Programming</i>	proficient: Claude, Python, PyTorch, SQL, Huggingface basic knowledge: Longchain, ChatGPT, Spark, C++, Keras, CUDA
<i>Language</i>	English, Mandarin
<i>Softskills</i>	collaboration, communication, vision
<i>Reviewer</i>	CVPR, AAAI, TMI, ICCV, MICCAI, CMIG, ICRA, IROS and so on

SELECTED PUBLICATIONS

Please kindly check my [Google scholar](#) for the full list.

- 1 . Hsieh Chen-I, Kang Zheng, Chihung Lin, Ling Mei, Le Lu, Weijian Li, Fang-Ping, Chen, Yirui Wang, **Xiao-Yun Zhou**, Fakai Wang, Guotong Xie, Jing Xiao, Shun Miao and Chang-Fu Kuo. Automated bone mineral density prediction and fracture risk assessment using plain radiographs via deep learning. *Nature communications*, 12(1), pp.1-9, 2021.
- 2 . Nanyang Ye, Jingxuan Tang, Huayu Deng, **Xiao-Yun Zhou**, Qianxiao Li and Zhenguo Li. Adversarial invariant learning. *CVPR*, pp.12441-12449, 2021.
- 3 . Nanyang Ye, Qianxiao Li, **Xiao-Yun Zhou** and Zhanxing Zhu. An Annealing Mechanism for Adversarial Training Acceleration. *TNNLS*, 2021.
- 4 . Yirui Wang, Kang Zheng, Chi-Tung Cheng, **Xiao-Yun Zhou**, Zhilin Zheng, Jing Xiao, Le Lu, Chien-Hung Liao and Shun Miao. Knowledge distillation with adaptive asymmetric label sharpening for semi-supervised fracture detection in chest x-rays. *IPMI*, pp.599-610, 2021.
- 5 . Ruoxi Wang, Dandan Zhang, Qingbiao Li, **Xiao-Yun Zhou** and Benny Lo. Real-time Surgical Environment Enhancement for Robot-Assisted Minimally Invasive Surgery Based on Super-Resolution. *ICRA*, pp.3434-3440, 2021.
- 6 . Nanyang Ye, Qianxiao Li, **Xiao-Yun Zhou** and Zhanxing Zhu. Amata: An Annealing Mechanism for Adversarial Training Acceleration. *AAAI*, pp.10691-10699, 2021.
- 7 . Yu Chen, Yuxuan Wang, Bolin Lai, Zijie Chen, Xu Cao, Nanyang Ye, Zhongyuan Ren, Junbo Zhao, **Xiao-Yun Zhou** and Peng Qi. VeniBot: Towards Autonomous Venipuncture with Semi-supervised Vein Segmentation from Ultrasound Images. accepted by *IROS* 2021.
- 8 . Kang Zheng, Yirui Wang, **Xiao-Yun Zhou**, Fakai Wang, Le Lu, Chihung Lin, Lingyun Huang, Guotong Xie, Jing Xiao, Chang-Fu Kuo and Shun Miao. Semi-Supervised Learning for Bone Mineral Density Estimation in Hip X-ray Images. *MICCAI*, pp.33-42, 2021.
- 9 . **Xiao-Yun Zhou**, Guang-Zhong Yang and Su-Lin Lee. A real-time and registration-free framework for dynamic shape instantiation. *MedIA*, 44: 86-97, 2018.
- 10 . **Xiao-Yun Zhou***, Zhao-Yang Wang*, Peichao Li, Jian-Qing Zheng and Guang-Zhong Yang. One-stage shape instantiation from a single 2D image to 3D point cloud. *MICCAI*, 30-38, 2019.
- 11 . **Xiao-Yun Zhou** and Guang-Zhong Yang. Normalization in training U-Net for 2D biomedical semantic segmentation. *RAL*, 4(2): 1792-1799 2019.
- 12 . **Xiao-Yun Zhou**, Celia Riga, Su-Lin Lee and Guang-Zhong Yang. Towards automatic 3D shape instantiation for deployed stent grafts: 2D multiple-class and class-imbalance marker segmentation with equally-weighted focal U-Net. *IROS*, 1261-1267, 2018.

- 13 . **Xiao-Yun Zhou**, Jianyu Lin, Celia Riga, Guang-Zhong Yang and Su-Lin Lee. Real-time 3D shape instantiation from single fluoroscopy projection for fenestrated stent graft deployment. *RAL*, 3(2): 1314-1321, 2018.
- 14 . Jian-Qing Zheng*, **Xiao-Yun Zhou***, Celia Riga and Guang-Zhong Yang. Real-time 3D shape instantiation of partially-deployed stent segment from a single 2D fluoroscopic image for fenestrated endovascular aortic repair. *RAL*, 4(4): 3703-3710, 2019.
- 15 . **Xiao-Yun Zhou***, Jian-Qing Zheng*, Peichao Li and Guang-Zhong Yang. ACNN: a full resolution DCNN for medical image segmentation. *ICRA*, 8455-8461, 2020.
- 16 . **Xiao-Yun Zhou**, Sabine Ernst and Su-Lin Lee. Path planning for robot-enhanced cardiac radiofrequency catheter ablation. *ICRA*, 4172-4177, 2016.
- 17 . Zhao-Yang Wang*, **Xiao-Yun Zhou***, Peichao Li and Guang-Zhong Yang. Instantiation-Net: 3D RV mesh instantiation from a single 2D MRI projection. *MICCAI*, 680-691, 2020.
- 18 . Peichao Li*, **Xiao-Yun Zhou***, Zhao-Yang Wang and Guang-Zhong Yang. Z-Net: an asymmetric 3D DCNN for medical CT volume segmentation. accepted by *IROS 2020*, pp 2906-2913, 2020.
- 19 . **Xiao-Yun Zhou**, Yao Guo, Mali Shen and Guang-Zhong Yang. Application of artificial intelligence in surgery. *Frontier of Medicine*, 1-14, 2020.
- 20 . Jian-Qing Zheng, **Xiao-Yun Zhou**, Celia Riga and Guang-Zhong Yang. Towards 3D path planning from a single 2D fluoroscopic image for robot assisted fenestrated endovascular aortic repair. *ICRA*, 8747-8753, 2019.
- 21 . Yingjing Feng, Ziyang Guo, Ziyang Dong, **Xiao-Yun Zhou**, Ka-Wai Kwok, Sabine Ernst and Su-Lin Lee. An efficient cardiac mapping strategy for radiofrequency catheter ablation with active learning. *IJCARS*, 12(7): 1199-1207, 2017.
- 22 . Bolin Lai, Yuhsuan Wu, Xiaoyu Bai, **Xiao-Yun Zhou**, Peng Wang, Jinzheng Cai, Yuankai Huo, Lingyun Huang, Yong Xia, Jing Xiao, Le Lu, Heping Hu and Adam Harrison. Liver Tumor Localization and Characterization from Multi-phase MR Volumes Using Key-Slice Prediction: A Physician-Inspired Approach. *MICCAI-PRIME*, pp. 47-58 2021.
- 23 . Bolin Lai, Yuhsuan Wu, **Xiao-Yun Zhou**, Peng Wang, Le Lu, Lingyun Huang, Mei Han, Jing Xiao, Heping Hu and Adam P Harrison. Hetero-Modal Learning and Expansive Consistency Constraints for Semi-Supervised Detection from Multi-Sequence Data. *MICCAI-MLMI* pp 296-305, 2021
- 24 . **Xiao-Yun Zhou**, Bolin Lai, Weijian Li, Yirui Wang, Kang Zheng, Fakai Wang, Chihung Lin, Le Lu, Lingyun Huang, Mei Han, Guotong Xie, Jing Xiao, Kuo Chang-Fu, Adam Harrison, Shun Miao. Scalable Semi-supervised Landmark Localization for X-ray Images using Few-shot Deep Adaptive Graph. *MICCAI-DALI* pp 145-153, 2021

* - equal contribution, J - journal, C - conference